

Aniyah McWilliams

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Project Experience

Know their names | Afro-American Historical Association of Fauquier County

May 2026-Present

- Collaborating with AAHA to analyze Albemarle County, Augusta County, and Fauquier County data to link enslaved individuals pre- and post- abolition of slavery to allow for tracing lineage
- Utilizing SPLINK and the Python Record Linkage Package to find matching individuals
- Analyzing census data, slave schedules, burial records, and more to create a composite score for the confidence of the same individual being mentioned across documents
- Developing an interactive dashboard with integrated search functionality by prioritizing user experience, visualizing record linkage confidence scores, and family trees

Interactive Career Dashboard | UVA School of Data Science

May 2026-Present

- Cleaned, standardized, and analyzed mixed-method survey data to produce comprehensive visualizations of post-graduate employment outcomes, including job locations, titles, and salaries
- Collaborated with Career Services and the Data Science department to balance project success and operational innovation while developing a Power BI Dashboard to display statistics

Amazon Beauty and Personal Product Review Analysis | Big Data Systems

May 2026

- Performed K-Means clustering on an 11 GB dataset consisting of 23.9 million Amazon Beauty and Personal Care product reviews
- Compared the runtime analysis across DuckDB and Apache Spark
- Detected five distinct clusters of Amazon reviews that revolved around skin products, hair products, nail products, perfumes and colognes, and makeup
- Discovered DuckDB had a quicker runtime than Apache Spark, but Apache Spark produced more cohesive and informative clusters

Exploration of Key Topics in Jet Magazine

Mar-April 2026

- Cleaned and analyzed 178 MB of Internet Archive Jet Magazine data from 1951 to 1978, removing stopwords, punctuation, and symbols introduced during page scanning
- Employed Sentiment Analysis, Principal Component Analysis, and Latent Dirichlet Allocation Topic Modeling in Python to explore cultural and social patterns throughout the magazines
- Discovered a shift in the cultural language and patterns in Jet Magazines produced before 1970 vs. after

Image Classification and Outfit Recommendation | Vision Vogue

Mar-April 2026

- Leveraged Convolutional Neural Network (CNN) models to classify images of clothing items based on their season, gender, and clothing article label
- Utilized PyTorch and TensorFlow to build ResNet-18, ResNet-50, and VGG-16 CNN models that used default weights and trained for 50 epochs on 1200 training images; evaluation was conducted using label accuracy on 300 test images
- Used the label predicted by the ResNet-18 model as input to a KNN model, which recommends the most similar clothing items from the other two clothing article labels to generate a complete outfit
- Generated outfit recommendations based on input image to highlight the potential of algorithms in outfit curation to minimize stress and effort

Prediction of Adverse Childhood Experiences and Adult Depression

Oct - Dec 2025

- Investigated the relationship between adverse childhood experiences (ACEs) and adult mental disorders, building Logistic Regression and KNN models with ~72% accuracy to predict adult depression
- Gathered and cleaned 5.85 MB of numerical and categorical data from the CDC's Behavioral Risk Factor Surveillance System, leveraging pandas to clean and concatenate the dataframes

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- Employed Python's Pandas and Sci-kit learn package to create captivating visualizations, execute profound analyses, and build Lasso Regression, Linear Regression, and Hierarchical Clustering Model
- Engineered an interactive dashboard that displayed key numerical summaries about our variables and evaluative metrics about the models
- Concluded that the data demonstrated two distinct groups of ACE exposure, very low and very high, and that the most predictive ACEs for adult disorders were: living with someone depressed, living with someone who swore at you, and divorced parents
- Utilized Python